



Replacing ground corn with soyhulls plus palmitic acid in low metabolizable protein diets with or without rumen-protected amino acids: Effects on production and nutrient utilization in dairy cows

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ABSTRACT

We previously observed that diets with reduced starch concentration decreased yields of milk and milk protein in dairy cows fed low metabolizable protein diets. Supplementation of reduced-starch diets with a lipid source may attenuate or eliminate production losses. Our objective was to investigate the effects of partially replacing ground corn with soyhulls plus a palmitic acid-enriched supplement on dry matter (DM) intake, milk yield and composition, plasma AA concentration, and N and energy utilization in cows fed low metabolizable protein diets (mean = −68 g/d balance) with or without rumen-protected Met, Lys, and His (RP-MLH). Sixteen multiparous Holstein cows averaging (mean ± standard deviation) 112 ± 28 d in milk, 724 ± 44 kg of body weight, and 46 ± 5 kg/d of milk in the beginning of the study were used in a replicated 4 × 4 Latin square design with a 2 × 2 factorial arrangement of treatments. Each period lasted 21 d, consisting of 14 d for diet adaptation and 7 d for data and sample collection. Diets were fed as follows: (1) high starch (HS), (2) HS plus RP-MLH (HS+AA), (3) reduced starch plus a palmitic acid-enriched supplement (RSPA), and (4) RSPA plus RP-MLH (RSPA+AA). The HS diet contained (DM basis) 26% ground corn and 7% soyhulls, and the RSPA diet had 10% ground corn, 22% soyhulls, and 1.5% palmitic acid. The HS diet averaged (DM basis) 32.6% starch and 4% ether extract, while starch

and ether extract concentrations of the RSPA diet were 21.7 and 5.9%, respectively. All 4 diets had (DM basis) 40% corn silage, 5% mixed-mostly grass haylage, 5% grass hay, and 50% concentrate. Diets did not affect DM intake and milk yield. Contrarily, feeding RSPA and RSPA+AA increased yields of energy-corrected milk (47.0 vs. 44.8 kg/d) and milk fat (1.65 vs. 1.50 kg/d) compared with HS and HS+AA. Milk fat concentration tended to decrease when RP-MLH was supplemented to HS, but no change was seen when added to RS (starch level × RP-MLH interaction). Milk and plasma urea N increased, and milk N efficiency decreased in cows fed RSPA and RSPA+AA versus HS and HS+AA. Apparent total-tract digestibilities of crude protein and neutral detergent fiber, as well as urinary urea N and total N excretion, were greater in cows offered RSPA and RSPA+AA than HS and HS+AA. Plasma Met and His concentrations increased with supplemental RP-MLH. Intake of gross energy and digestible energy and the output of urinary and milk energy were all greater with feeding RSPA and RSPA+AA versus HS and HS+AA. In summary, partially replacing ground corn with soyhulls plus palmitic acid in diets supplemented or not with RP-MLH increased milk fat yield and fiber digestibility and maintained DM intake and milk yield, but with decreased milk N efficiency and elevated urinary N excretion.

Key words: fatty acid, methane, milk yield, starch

INTRODUCTION

Several studies have compared the effects of high-starch (i.e., glucogenic) versus high-fat (i.e., lipogenic) diets on production performance and nutrient utilization in dairy cows (e.g., van Knegsel et al., 2007; Boerman et al., 2015; Morris et al., 2020). In general, glucogenic diets partition dietary energy into body reserves, whereas lipogenic diets promote milk fat synthesis (van

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Knegsel et al., 2007; Boerman et al., 2015). Boerman et al. (2015) concluded that these differences in dietary energy use may be explained by the synergistic effects of increased insulin concentration and elevated supply of ruminal biohydrogenation intermediates in cows fed glucogenic diets, thereby prioritizing nutrient storage at expense of mammary fat production. However, information on whether or how glucogenic or lipogenic sources may interact with rumen-protected (RP)-AA to modulate production responses and nutrient utilization in cows is lacking. Currently, feeding low MP diets is one of the most effective strategies to reduce urinary N excretion (Lee et al., 2012), but may come at the expense of milk and milk protein production (Lee et al., 2012; Giallongo et al., 2016). Supplementation of low MP diets with RP Met, Lys, and His (RP-MLH) alleviated production losses (Lee et al., 2012; Giallongo et al., 2016) and improved N utilization (Lee et al., 2012). Taken together, investigating the interplay between energy source (i.e., glucogenic vs. lipogenic) and RP-AA is key for understanding metabolic processes underpinning milk yield and nutrient utilization in dairy cows fed low MP diets. Nichols et al. (2019) reported significant dietary protein level $\times$ supplemental fat (hydrogenated palm fatty acids; 50% palmitic acid and 47% stearic acid) interactions for the plasma concentrations of individual (i.e., His, Ile, Leu, Val) and total EAA. Specifically, energy from protein (50:50 xylose-treated soybean meal/rapeseed meal mix) increased arterial concentration of EAA to a greater magnitude in the low (+56.7%) than the high-fat (+28.3%) diet, indicating that fat supplementation possibly affected intestinal absorption of EAA or their utilization by extramammary tissues (Nichols et al., 2019).

In our previous research (Zang et al., 2021b), we investigated the effects of high-starch (34.4% starch; 30% ground corn) versus reduced-starch (12.3% starch; 20% pelleted beet pulp plus 10% soyhulls) diets supplemented or not with RP-MLH on production performance and nutrient utilization in dairy cows fed low MP diets. However, no interactions were observed for most variables measured by Zang et al. (2021b). Nevertheless, partially replacing ground corn with soyhulls and a palmitic acid-enriched supplement in low MP diets with or without RP-MLH could result in nutrient interactions due to the potential effect of supplemental fat on reducing the supply of EAA to mammary tissues as reported by Nichols et al. (2019), thus warranting further research.

We hypothesized that partially replacing energy from ground corn (glucogenic substrate) with energy from soyhulls and palmitic acid (lipogenic substrates) would lead to interactions with RP-MLH supplementation, which could ultimately affect nutrient utilization in

dairy cows fed low MP diets formulated to be isocaloric and isonitrogenous. We further hypothesized that energy from soyhulls and supplemental palmitic acid would compensate energy from ground corn resulting in similar milk yield but greater milk fat yield. Our objective was to investigate the effects of feeding soyhulls plus a palmitic acid-enriched supplement at expense of ground corn on production performance, apparent total-tract digestibility of nutrients, urinary excretion of N, plasma concentration of AA, and energy utilization in dairy cows fed low MP diets with or without RP-MLH.

MATERIALS AND METHODS

All experimental procedures were approved by the Institutional Animal Care and Use Committee (protocol no. 190202) of the University of New Hampshire (Durham). The study was carried out at the University of New Hampshire Fairchild Dairy Teaching and Research Center (Durham) from March 11 to June 9, 2019.

Cows, Experimental Design, and Treatments

Sixteen multiparous Holstein cows averaging (mean $\pm$ SD) 112 ± 28 DIM, 46 ± 5 kg/d of milk, and 724 ± 44 kg of BW at the beginning of the study were enrolled. Cows were housed in a tie-stall barn equipped with feed tubs for individualized feed intake and water bowls for ad libitum access to water. Cows were milked twice daily at 0530 and 1630 h, with milk yield recorded at every milking over the duration of the experiment. Body weight was taken for 3 consecutive days immediately after the afternoon milking before the beginning of the experiment and at the end of each experimental period to compute BW change. Body condition score was assessed by 3 trained individuals before the beginning of the study and on the last day of each experimental period following the procedures described by Wildman et al. (1982). Dietary ingredients were mixed and offered as TMR twice per day at 0600 and 1700 h using a Super Data Ranger mixer (American Calan Inc.). Refusals were collected and weighed daily before the afternoon feeding. Feed offered was adjusted daily to allow for 5 to 10% refusals, with individual feed intake recorded throughout the study.

Cows were blocked by DIM and milk yield and, within each block, assigned randomly to treatment sequences in a replicated 4×4 Latin square design with a 2×2 factorial arrangement of treatments. Squares were balanced for potential first-order carryover effects in subsequent periods as each treatment immediately preceded and followed each other once in individual

Table 1. Nutrient composition of ingredients (mean $\pm$ SD) used in the experimental diets (% of DM, unless otherwise noted)

Item	Corn silage	Haylage ¹	Grass hay	Ground corn	Soyhulls	Protein blend ²	Soybean meal	Urea
No. of samples	4	4	4	4	4	4	4	4
DM, % of fresh matter	33.1 $\pm$ 1.57	38.6 $\pm$ 0.76	89.8 $\pm$ 0.74	89.2 $\pm$ 1.15	91.7 $\pm$ 1.69	90.1 $\pm$ 0.67	91.3 $\pm$ 0.80	99.7 $\pm$ 0.17
CP	7.93 $\pm$ 0.05	16.5 $\pm$ 0.78	9.35 $\pm$ 0.66	8.80 $\pm$ 0.08	17.5 $\pm$ 1.27	43.6 $\pm$ 0.21	52.6 $\pm$ 0.29	290 $\pm$ 1.82
NDF	41.8 $\pm$ 1.35	54.4 $\pm$ 1.45	67.3 $\pm$ 1.61	8.60 $\pm$ 0.61	53.4 $\pm$ 2.51	24.8 $\pm$ 0.50	10.8 $\pm$ 0.96	NA ³
ADF	24.7 $\pm$ 1.12	36.7 $\pm$ 1.26	42.2 $\pm$ 0.94	2.33 $\pm$ 0.26	39.2 $\pm$ 2.10	17.6 $\pm$ 0.75	7.85 $\pm$ 0.66	NA
ADL	4.15 $\pm$ 0.74	7.20 $\pm$ 0.66	6.28 $\pm$ 0.50	1.05 $\pm$ 0.37	3.60 $\pm$ 1.55	7.38 $\pm$ 0.53	2.25 $\pm$ 1.57	NA
Starch	34.3 $\pm$ 1.00	1.75 $\pm$ 0.30	0.50 $\pm$ 0.29	70.3 $\pm$ 1.83	2.95 $\pm$ 1.27	1.43 $\pm$ 0.30	1.23 $\pm$ 0.17	NA
Ether extract	3.70 $\pm$ 0.14	4.75 $\pm$ 0.26	1.95 $\pm$ 0.24	4.20 $\pm$ 0.22	7.05 $\pm$ 0.85	6.15 $\pm$ 0.54	1.30 $\pm$ 0.14	NA
NE _L , Mcal/kg of DM	1.62 $\pm$ 0.05	1.28 $\pm$ 0.03	1.03 $\pm$ 0.03	2.10 $\pm$ 0.01	1.72 $\pm$ 0.11	1.72 $\pm$ 0.04	1.78 $\pm$ 0.03	NA
Ash	3.86 $\pm$ 0.20	8.49 $\pm$ 0.82	4.65 $\pm$ 0.46	1.22 $\pm$ 0.43	5.09 $\pm$ 1.01	8.10 $\pm$ 0.73	7.94 $\pm$ 0.57	NA
Ca	0.17 $\pm$ 0.01	0.82 $\pm$ 0.04	0.40 $\pm$ 0.04	0.01 $\pm$ 0.01	0.51 $\pm$ 0.03	0.67 $\pm$ 0.01	0.58 $\pm$ 0.05	NA
P	0.32 $\pm$ 0.01	0.35 $\pm$ 0.03	0.23 $\pm$ 0.01	0.32 $\pm$ 0.01	0.26 $\pm$ 0.01	1.00 $\pm$ 0.02	0.80 $\pm$ 0.02	NA
Mg	0.16 $\pm$ 0.01	0.27 $\pm$ 0.03	0.19 $\pm$ 0.02	0.11 $\pm$ 0.01	0.27 $\pm$ 0.01	0.52 $\pm$ 0.01	0.33 $\pm$ 0.01	NA
K	1.10 $\pm$ 0.05	2.56 $\pm$ 0.25	1.60 $\pm$ 0.20	0.40 $\pm$ 0.02	1.50 $\pm$ 0.10	1.74 $\pm$ 0.03	2.44 $\pm$ 0.05	NA
Na	0.01 $\pm$ 0.00	0.06 $\pm$ 0.02	0.09 $\pm$ 0.01	0.00 $\pm$ 0.00	0.01 $\pm$ 0.00	0.10 $\pm$ 0.01	0.01 $\pm$ 0.01	NA
S	0.11 $\pm$ 0.00	0.24 $\pm$ 0.02	0.15 $\pm$ 0.01	0.10 $\pm$ 0.00	0.18 $\pm$ 0.01	0.82 $\pm$ 0.02	0.43 $\pm$ 0.01	NA
Fe, mg/kg of DM	141 $\pm$ 20.7	248 $\pm$ 76.5	94.3 $\pm$ 20.5	40.0 $\pm$ 5.23	399 $\pm$ 25.6	177 $\pm$ 7.62	90.8 $\pm$ 6.65	NA
Zn, mg/kg of DM	22.5 $\pm$ 1.29	27.5 $\pm$ 1.73	25.8 $\pm$ 1.26	21.3 $\pm$ 0.96	51.8 $\pm$ 2.22	58.5 $\pm$ 1.73	49.0 $\pm$ 0.82	NA
Cu, mg/kg of DM	5.00 $\pm$ 2.00	5.25 $\pm$ 0.50	5.00 $\pm$ 0.82	0.00 $\pm$ 0.00	6.25 $\pm$ 0.50	8.50 $\pm$ 0.58	13.3 $\pm$ 0.50	NA
Mn, mg/kg of DM	13.5 $\pm$ 1.29	34.8 $\pm$ 4.99	49.8 $\pm$ 4.65	5.00 $\pm$ 0.00	19.3 $\pm$ 0.96	55.3 $\pm$ 1.26	36.3 $\pm$ 1.26	NA
Mo, mg/kg of DM	0.80 $\pm$ 0.00	4.85 $\pm$ 0.68	2.03 $\pm$ 0.13	0.70 $\pm$ 0.29	1.70 $\pm$ 0.54	3.08 $\pm$ 0.25	4.75 $\pm$ 0.42	NA

¹Haylage = mixed-mostly grass haylage.²Protein blend = plant-based ruminally protected protein supplement containing canola meal and soybean meal (AminoMax; Afgriotech LLC).³NA = not analyzed.

squares (Williams, 1949). Each experimental period lasted 21 d, with the first 14 d used for diet adaptation and the last 7 d for data and sample collection. Dietary treatments were (1) high-starch diet (**HS**), (2) HS plus RP-MLH (**HS+AA**), (3) reduced-starch diet plus a palmitic acid-enriched supplement (**RSPA**); and (4) RSPA plus RP-MLH (**RSPA+AA**). The basal diets were formulated (NRC, 2001) to be isocaloric and isonitrogenous for meeting the nutrient requirements, except MP, of a lactating dairy cow averaging 120 DIM, weighing 700 kg of BW, consuming 26 kg/d of DM, and producing 42 kg/d of milk with 3.5% fat, 3.1% true protein, and 4.98% lactose. All diets contained (DM basis) 40% corn silage, 5% mixed-mostly grass haylage, 5% grass hay, and 50% concentrate. The HS basal diet had (DM basis) 26% ground corn, and the RSPA basal diet averaged 10% ground corn, 22% soyhulls, and 1.5% palmitic acid-enriched supplement (BergaFat F100; Berg+Schmidt America LLC). Based on the manufacturer's specification, the palmitic acid-enriched supplement contains a minimum of 80% palmitic acid. Twelve, 9, and 15 g/d of RP-Met (Smartamine M; Adisseo USA Inc.), RP-Lys (AjiPro-L; Ajinomoto Health & Nutrition North America Inc.), and RP-His (Ajinomoto prototype supplement; Ajinomoto Co. Inc.), respectively, were top-dressed to the TMR and offered to cows. The RP-Met, RP-Lys, and RP-His supplements contained 75% DL-Met with 80% bioavailability (Chirgwin et al., 2015), 40% Lys with 54% bioavailability (Giallongo et al., 2016), and 40% His with 49% bioavailability (ac-

cording to the manufacturer) and were expected to deliver 7, 2, and 3 g/d of digestible Met, Lys, and His, respectively. The nutrient and AA composition of individual ingredients used in the experimental diets are presented in Tables 1 and 2, respectively. The ingredient and nutrient composition of the 2 basal diets are reported in Table 3, and the NRC (2001) evaluation of all 4 diets is shown in Table 4. The nutrient (Table 1) and AA (Table 2) composition of feeds used in the experimental diets, as well as actual animal variables (i.e., DIM, lactation number, and BW), DMI, milk yield, and concentrations of milk fat, true protein, and lactose, were used in the NRC (2001) software to obtain the estimates reported in Table 4.

Feed Sampling and Analyses

Samples of corn silage, mixed-mostly grass haylage, grass hay, TMR, and refusals were collected thrice per week and composited on a weekly basis. Samples of concentrates (i.e., ground corn, soyhulls, protein blend, and soybean meal) were taken once per week. Feed samples were dried for 48 h at 55°C in a forced-air oven (VWR Scientific) for determination of DM to adjust the TMR, on an as-fed basis, and to calculate DMI over the duration of the study. Weekly samples of dietary ingredients were lyophilized (Labconco Inc.), ground to pass through a 1-mm screen using a Wiley mill (A. H. Thomas Co.), and finally stored in air-tight glass jars until shipped for nutrient analyses.

Table 2. Amino acid composition of ingredients used in the experimental diets (n = 1 composite sample per feedstuff)

Item	Corn silage	Haylage ¹	Grass hay	Ground corn	Soyhulls	Protein blend ²	Soybean meal
Total AA, g/100 g	5.31	11.3	6.34	8.83	15.7	36.4	47.2
EAA, % of total AA							
Arg	1.88	3.71	4.73	4.53	6.13	6.56	7.27
His	1.32	1.77	1.89	2.83	2.68	2.77	2.67
Ile	4.71	5.30	4.89	3.74	4.66	4.67	4.83
Leu	12.1	9.10	8.83	12.1	7.59	7.74	7.91
Lys	2.45	5.21	5.68	3.51	6.96	5.90	6.51
Met	2.07	1.94	1.89	2.04	1.40	1.89	1.42
Phe	4.52	5.83	5.68	4.98	4.85	4.77	5.32
Thr	3.20	4.77	4.89	3.51	3.83	4.31	3.90
Trp	0.56	0.97	0.95	0.68	0.89	1.37	1.46
Val	6.21	6.89	6.47	4.98	4.98	5.41	4.98
NEAA, % of total AA							
Ala	13.2	8.66	7.10	7.47	4.59	4.67	4.41
Asp	5.27	9.72	10.3	7.02	10.5	9.08	11.3
Cys	1.69	1.15	1.42	2.27	1.79	2.31	1.46
Gly	5.27	5.74	5.68	3.96	6.38	5.13	4.32
Glu	12.8	9.81	11.4	18.5	15.1	18.4	18.2
Hydroxylysine	6.97	3.53	1.89	0.23	0.45	0.38	0.17
Hydroxyproline	0.56	1.33	1.10	0.23	2.17	0.63	0.17
Orn	0.19	0.62	0.16	0.11	0.06	0.05	0.08
Pro	7.91	5.65	6.15	8.83	5.36	6.17	5.15
Ser	3.01	3.80	4.26	4.53	4.72	4.12	4.30
Try	2.07	3.27	2.52	2.72	3.83	3.43	3.92
Taurine	2.07	1.06	2.05	1.25	1.08	0.25	0.19

¹Haylage = mixed-mostly grass haylage.²Protein blend = plant-based ruminally protected protein supplement containing canola meal and soybean meal (AminoMax; Afgriotech LLC).

Lyophilized ground samples of individual ingredients were shipped to Dairy One Forage Laboratory (Ithaca, NY) and analyzed for DM (method 930.15; AOAC International, 2016), CP (total N $\times$ 6.25; method 990.03; AOAC International, 2016), soluble CP (Krishnamoorthy et al., 1982), α -amylase, sodium sulfite-treated NDF (method 6, Ankom Technology; solutions as in Van Soest et al., 1991), ADF [method 5, Ankom Technology; solutions as in method 973.18 (AOAC International, 2016)], ADL [method 9, Ankom Technology in a Ankom Daisy Incubator; solutions as in method 973.18 (AOAC International, 2016)], starch (YSI 2700 Select Biochemistry Analyzer, application note no. 319; YSI Inc. Life Sciences), ether extract [extraction by a Soxtec HT6 System (Foss North America) using anhydrous diethyl ether; method 2003.05 (AOAC International, 2016)], and ash (method 942.05; AOAC International, 2016). Individual minerals were analyzed using an iCAP 6300 Intrepid Inductively Coupled Plasma Radial Spectrometer (Thermo Fisher Scientific Inc.) after microwave digestion. Additionally, TMR and refusals were analyzed for CP, NDF, ADF, ash, and gross energy (**GE**; IKA C2000 basic calorimeter system; KA Works Inc.) at Dairy One Forage Laboratory. Weekly samples of individual ingredients were composited over the 12-wk study, ground to pass through a 0.5-mm screen (Willey grinder; A. H. Thomas Co.),

and shipped to the University of Missouri Agricultural Experiment Station Chemical Laboratory (Columbia, MO) for AA analyses by cation exchange chromatography coupled with postcolumn ninhydrin derivatization using norleucine as the internal standard (method 982.30; AOAC International, 2016). Tryptophan was measured after alkaline hydrolysis and sulfur AA were determined following performic acid oxidation (method 988.15; AOAC International, 2016).

Milk and Blood Sampling and Analyses

Milk samples were collected using automatic samplers during 4 consecutive milkings, beginning in the afternoon milking of d 15 of each experimental period and transferred into 50-mL tubes preserved with 2-bromo-2-nitropropane-1,3 diol (Broad Spectrum Microtabs II; Advanced Instruments Inc.). Samples were stored at 4°C until shipped overnight in cold ice packs to Dairy One DHIA (Ithaca, NY) for analyses of fat, true protein, lactose, and MUN by Fourier-transform infrared spectroscopy using a MilkoScan FT+ (Foss Inc.).

Blood samples were collected using vacutainer 15% EDTA tubes (Monoject) via puncture of the coccygeal blood vessels approximately 4 h after the morning feedings on d 16 and 17 of each experimental period, with tubes immediately placed in a chill bucket with beads

Table 3. Ingredient and nutrient composition (% of DM, unless otherwise noted) of the basal diets used in the experiment

Item	Basal diet ¹	
	HS	RSPA
Ingredient		
Corn silage	40.0	40.0
Mixed-mostly grass haylage	5.00	5.00
Grass hay	5.00	5.00
Ground corn	26.1	10.0
Soyhulls	7.04	22.0
Protein blend ²	8.00	8.01
Soybean meal	5.81	5.59
Mineral and vitamin premix ³	2.00	2.00
BergaFat F100 ⁴	—	1.50
Urea	0.54	0.46
Sodium bicarbonate	0.50	0.50
Nutrient composition		
DM, % of fresh matter	46.7	47.0
CP	15.9	16.8
NDF	31.4	38.0
Forage NDF	22.8	22.8
ADF	19.1	24.5
Starch	32.6	21.7
Ether extract	4.00	5.90
NE _L , ⁵ Mcal/kg of DM	1.56	1.61
Ash	3.98	4.53
Ca	0.60	0.60
P	0.40	0.40
Digestible His, ⁵ % of MP	2.07	2.15
Digestible Met, ⁵ % of MP	1.84	1.86
Digestible Lys, ⁵ % of MP	6.38	6.74
Gross energy, ⁵ Mcal/kg of DM	4.14	4.24

¹HS = high-starch diet; RSPA = reduced starch + palmitic acid-enriched supplement.

²Protein blend = plant-based ruminally protected protein supplement containing canola meal and soybean meal (AminoMax; Afgritech LLC).

³Mineral and vitamin premix provided (DM basis): 14.8% Ca, 1.18% P, 5.45% Mg, 8.53% Cl, 0.10% K, 14.1% Na, 0.44% S, 45 mg/kg Co, 396 mg/kg Cu, 2,150 mg/kg Fe, 970 mg/kg Mn, 1,350 mg/kg Zn, 2,500 kIU/kg vitamin A, 400 kIU/kg vitamin D, and 1 kIU/kg vitamin E.

⁴BergaFat F100 (Berg+Schmidt America LLC) is a ruminally stable lipid supplement containing 80% palmitic acid.

⁵Estimated using the NRC (2001) model with actual animal variables (DIM, lactation number, and BW), DMI, milk yield and composition, and nutrient and AA composition of dietary ingredients used during the experiment.

(Chemglass Life Sciences). Samples were then transported to the laboratory and centrifuged (Eppendorf model 5810) at $2,155 \times g$ for 20 min at 4°C. Plasma samples were pooled over the 2 d by cow and period, with composited plasma analyzed for the concentrations of AA, carnosine, and urea N (**PUN**) by Ajinomoto Co. Inc. (Kawasaki-shi, Japan) using a High-Speed AA Analyzer L-8900 (Hitachi High-Technologies Co.) following the procedures described by the manufacturer (<https://www.hitachi-hightech.com/us/library/literature/brochure-l-8900-amino-acid-analyzer.html>; accessed February 10, 2022). Plasma samples were codified and shipped to Ajinomoto Co. Inc. to preserve treatment identity.

Table 4. NRC (2001) evaluation of the experimental diets containing different starch concentrations supplemented or not with rumen-protected Met, Lys, and His (RP-MLH)¹

Item ²	Diet ³			
	HS	HS+AA	RSPA	RSPA+AA
NE _L , Mcal/d				
Requirement	42.9	42.9	44.1	43.9
Supply	44.2	45.1	46.7	45.8
Balance	1.3	2.2	2.6	1.9
MP, g/d				
Requirement	3,086	3,154	3,115	3,069
Supply	2,964	3,040	3,118	3,037
Balance	−122	−113	4	−32
RDP, g/d				
Requirement	2,793	2,852	2,890	2,803
Supply	2,895	2,958	3,075	3,014
Balance	102	106	185	185
RUP, g/d				
Requirement	1,765	1,804	1,798	1,787
Supply	1,615	1,664	1,803	1,746
Balance	−150	−139	5	−41
Digestible His, g/d				
Requirement ⁴	68	69	69	68
Supply from the diet	64	65	67	65
Supply from RP-His	0	3	0	3
Balance	−4	−1	−2	0
Digestible Met, g/d				
Requirement ⁴	68	69	69	68
Supply from the diet	57	58	58	57
Supply from RP-Met	0	7	0	7
Balance	−11	−4	−11	−4
Digestible Lys, g/d				
Requirement ⁴	204	207	207	204
Supply	197	201	210	205
Supply from RP-Lys	0	2	0	2
Balance	−7	−4	3	3

¹RP-MLH = 12 g/d of RP-Met (Smartamine M; Adisseo USA Inc.), 9 g/d of RP-Lys (AjiPro-L; Ajinomoto Health & Nutrition North America Inc.), and 15 g/d of RP-His (prototype supplement; Ajinomoto Co. Inc.); RP-Met, RP-Lys, and RP-His supplements contained 75% DL-Met with 80% bioavailability (Chirgwin et al., 2015), 40% Lys with 54% bioavailability (Giallongo et al., 2016), and 40% His with 49% bioavailability (according to the manufacturer).

²All values were estimated using the NRC (2001) model with actual animal variables (DIM, lactation number, and BW), DMI, milk yield and composition, and nutrient and AA composition of dietary ingredients used in the experiment.

³HS = high-starch diet; HS+AA = HS + RP-MLH; RSPA = reduced starch + palmitic acid-enriched supplement; and RSPA+AA = reduced starch + palmitic acid-enriched supplement + RP-MLH; BergaFat F100 (Berg+Schmidt America LLC), which is a ruminally stable lipid supplement containing 80% palmitic acid, was fed at 1.5% of the diet DM.

⁴Requirements of digestible His, Met, and Lys were calculated as 2.2, 2.2, and 6.6% of MP requirements, respectively (Schwab et al., 2005).

Fecal and Urinary Sampling and Analyses

Fecal grab samples were collected directly from the rectum or during voluntary defecation across 8 time points over 3 d of each experimental period as follows: 0600 and 1500 h (d 18), 0300, 0900, and 1800 h (d 19), and 0000, 1200, and 2100 h (d 20). Approximately 200 g of fecal samples were taken during each sampling,

transferred into 4-L storage bags to obtain a composite sample (wet weight) by cow/period, and stored at -20°C until further processing. Samples were thawed at room temperature, placed in aluminum trays, dried (55°C , ~ 72 h) in a forced-air oven (VWR Scientific), and ground to pass through a 1-mm screen (Wiley mill; A. H. Thomas Co.). Dried fecal samples were shipped to Dairy One Forage Laboratory for wet chemistry analyses of DM, CP, NDF, starch, ether extract, ash, and GE as done for feeds. Triplicate samples (~ 0.5 g) of TMR, refusals, and feces were weighed into Ankom F57 bags ($25\text{ }\mu\text{m}$ pore size; Ankom Technology) and placed in a larger laundry nylon bag before inserting in the rumen of 1 ruminally cannulated late-lactation Holstein cow for 12 d. This cow was fed a corn silage-grass silage-based TMR with a 60:40 forage-to-concentrate ratio. Following removal from the rumen, bags were rinsed with tap water and analyzed in-house for NDF using an Ankom²⁰⁰⁰ fiber analyzer [Ankom Technology method 6; solution as in Van Soest et al. (1991)]. Indigestible NDF was used as the internal marker to estimate fecal output of DM and apparent total-tract digestibility of nutrients (Cochran et al., 1986; Huhtanen et al., 1994).

Spot samples of urine (~ 100 mL) were collected concurrently with feces via stimulation of the pudendal nerve by massaging the area below the vulva or during a voluntary urination event. After each sampling, 1 mL of urine (total = 8 mL over 3 d) was pipetted into 50-mL tubes containing 32 mL of $0.072\text{ N H}_2\text{SO}_4$ and kept refrigerated at 4°C until sampling was completed (d 18 to 20). Composite samples of urine obtained by cow per experimental period were stored at -20°C before analyses of nitrogenous metabolites. Samples were thawed inside a refrigerator (4°C) and analyzed for creatinine (assay kit no. 500701; Cayman Chemical Co.) using a chromate microplate reader set at a wavelength of 492 nm (Awareness Technology Inc.), urea N (Stanbio Urea Nitrogen Kit 580; Stanbio Laboratory Inc.) with a UV/visible spectrophotometer (Beckman Coulter Inc.) set at a wavelength of 520 nm, and total N [micro-Kjeldahl analysis (AOAC, 1990); Dairy One Forage Laboratory]. Daily urinary volume was estimated assuming a constant creatinine excretion rate of 29 mg/kg of BW (Valadares et al., 1999). Body weight measured in the last 3 d of each experimental period was used to estimate urinary volume. Urinary excretion of urea N and total N were calculated by multiplying their concentrations in urine by the urinary volume.

Gaseous Emission Measurements

Emissions of CO_2 and enteric CH_4 were measured at 0200 and 1400 h (d 15), 0500 and 1700 (d 16), 0800 and 2000 h (d 17), and 1100 and 2300 h (d 18) using

a GreenFeed unit (C-Lock Inc.), which was placed in front of each cow for approximately 5 min to sample breath and eructated gases, and then moved to the barn alley for 2 min to sample background gases. The unit was moved from cow to cow sequentially taking about 2 h to complete a sampling event (mean = 5:07 min/cow). Cows were trained to access the GreenFeed unit for 2 wk before the beginning of the experiment. A soybean meal-nonforage fiber source-based bait pellet (Hi-Line 16% Dairy/Beef Pellet; Poulin Grain Inc.) containing 19.6% CP, 35% NDF, 15.2% ADF, 16.7% starch, 6.2% ether extract, and 4.56 Mcal/kg of GE was used to attract cows to the unit. Approximately 25 g of bait pellet (as-fed basis) were dropped every 15 s leading to 0.44 kg of bait DMI per cow/sampling point that was added to TMR DMI to compute total DMI. A complete description of the gaseous sampling protocols and emission calculations was reported previously (Dorich et al., 2015).

Calculations

Yields of milk fat, true protein, and lactose were calculated using milk yield and concentrations of these milk components obtained from samples collected during d 15 to 17 of each experimental period. Intake of digestible energy (DE) and ME were calculated as follows:

$$\text{DE intake (Mcal/d)} =$$

$$\text{GE intake (Mcal/d)} - \text{fecal energy (Mcal/d)}.$$

$$\text{ME intake (Mcal/d)} = \text{DE intake (Mcal/d)}$$

$$- \text{urinary energy (Mcal/d)} - \text{CH}_4 \text{ energy (Mcal/d)}.$$

Urinary energy was estimated using Morris et al. (2021) equation as follows: urinary energy (Mcal/d) = $[14.6 \times \text{urinary N excretion (g/d)}]/1,000$. Methane energy was calculated as CH_4 production (L/d) multiplied by CH_4 enthalpy (9.45 kcal/L). Milk energy was estimated using the NASEM (2021) equation: milk energy (Mcal/d) = $[(0.0929 \times \text{milk fat}\%) + (0.0585 \times \text{milk true protein}\%) + (0.0395 \times \text{milk lactose}\%)] \times \text{milk yield (kg/d)}$.

Statistical Analyses

Data were analyzed using the MIXED procedure of SAS (version 9.4; SAS Institute Inc.) according to the following model:

$$Y_{ijklm} = \mu + S_i + C_{j(i)} + P_k + SC_l + \text{RP-MLH}_m + SC_l \times \text{RP-MLH}_m + e_{ijklm},$$

Table 5. Dry matter intake, milk yield and composition, MUN and plasma urea N (PUN) concentrations, BCS, and BW in lactating dairy cows fed low MP diets with different starch concentrations supplemented or not with rumen-protected Met, Lys, and His (RP-MLH)¹

Item	Diet ²				SEM	P-value ³		
	HS	HS+AA	RSPA	RSPA+AA		SC	MLH	SC × MLH
DMI, kg/d	28.3	29.0	29.1	28.4	0.75	0.76	0.92	0.12
Starch intake, kg/d	9.20	9.44	6.30	6.16	0.21	<0.001	0.66	0.13
Milk yield, kg/d	44.6	45.0	45.6	44.8	1.18	0.17	0.52	0.07
Milk yield/DMI, kg/kg	1.58	1.55	1.58	1.61	0.04	0.33	0.85	0.24
4% FCM, ⁴ kg/d	40.1	40.0	42.7	43.0	1.32	<0.001	0.84	0.70
4% FCM/DMI, kg/kg	1.42	1.38	1.48	1.52	0.03	<0.001	0.87	0.07
ECM, ⁵ kg/d	45.0	44.6	46.8	47.1	14.0	<0.001	0.96	0.45
ECM/DMI, kg/kg	1.57	1.54	1.62	1.66	0.03	<0.001	0.67	0.10
Milk fat, %	3.40	3.29	3.59	3.66	0.11	<0.001	0.62	0.04
Milk fat, kg/d	1.52	1.47	1.64	1.65	0.06	<0.001	0.48	0.15
Milk true protein, %	3.15	3.19	3.09	3.11	0.03	<0.001	0.06	0.85
Milk true protein, kg/d	1.41	1.42	1.40	1.41	0.04	0.65	0.36	0.82
Milk lactose, %	4.99	4.99	4.95	4.93	0.03	<0.01	0.49	0.26
Milk lactose, kg/d	2.22	2.24	2.26	2.23	0.06	0.53	0.85	0.46
Milk SCC, × 1,000 cells/mL	47.9	45.6	45.2	53.0	15.7	0.68	0.63	0.35
Milk N, % of N intake	30.4	30.5	29.1	29.4	0.65	<0.01	0.64	0.81
MUN, mg/dL	12.0	12.2	14.7	15.0	0.41	<0.001	0.33	0.90
PUN, mg/dL	12.4	12.1	15.4	15.7	0.46	<0.001	1.00	0.29
BCS	3.13	3.14	3.08	3.11	0.09	0.15	0.39	0.70
BCS change, point/21 d	0.09	0.13	0.08	0.06	0.03	0.20	0.69	0.37
BW, kg	776	779	773	775	12.3	0.16	0.20	0.73
BW change, kg/d	0.69	0.96	0.52	0.76	0.15	0.21	0.09	0.91

¹RP-MLH = 12 g/d of RP-Met (Smartamine M; Adisseo USA Inc.), 9 g/d of RP-Lys (AjiPro-L; Ajinomoto Health & Nutrition North America Inc.), and 15 g/d of RP-His (prototype supplement; Ajinomoto Co. Inc.).

²HS = high-starch diet; HS+AA = HS + RP-MLH; RSPA = reduced starch + palmitic acid-enriched supplement; and RSPA+AA = reduced starch + palmitic acid-enriched supplement + RP-MLH; BergaFat F100 (Berg+Schmidt America LLC), which is a ruminally stable lipid supplement containing 80% palmitic acid, was fed at 1.5% of the diet DM.

³SC = main effect of dietary starch concentration; MLH = main effect of RP-MLH supplementation; and SC × MLH = interaction between dietary starch concentration and RP-MLH supplementation.

⁴4% FCM = (0.4 × kg of milk) + (15 × kg of milk fat); Gaines and Davidson (1923).

⁵ECM = (0.327 × kg of milk) + (12.95 × kg of milk fat) + (7.65 × kg of milk protein); Tyrrell and Reid (1965).

where Y_{ijklm} = dependent variable, μ = overall mean, S_i = fixed effect of square ($i = 1$ to 4), $C_{j(i)}$ = random effect of cow ($j = 1$ to 16) nested within square, P_k = fixed effect of period ($k = 1$ to 4), SC_l = fixed effect of dietary starch concentration ($l = \text{high vs. reduced}$), $RP\text{-}MLH_m$ = fixed effect of RP-MLH supplementation ($m = \text{yes vs. no}$), $SC_l \times RP\text{-}MLH_m$ = interaction between dietary starch concentration and RP-MLH supplementation, and e_{ijklm} = residual error. Normality of residuals and homogeneity of variances were checked with normal probability and box plots and plots of residual versus predicted values, respectively. All results are reported as LSM and SEM, with the greatest SEM values shown in Tables 5, 6, 7, and 8. The main effects of dietary starch concentration and RP-MLH supplementation, and their interactions, were tested using ANOVA. The Tukey test was used to separate means when the interaction was significant (i.e., milk fat concentration). Significance was declared at $P \leq 0.05$ and tendencies at $0.05 < P \leq 0.10$.

Supplemental Table S1 (https://figshare.com/articles/online_resource/Supplemental_Tables_S1-3/22422811; Zang, 2023) shows the number of observations and the number of outliers removed (or not) for each variable response used in the statistical analyses. Outliers were removed from statistical analyses when studentized residuals were >3.0 or <-3.0 using the SORT procedure of SAS (version 9.4). One cow had decreased DMI and milk yield during period 1 and recovered thereafter, but all her data while she was apparently sick were removed from the statistical analyses. Another cow did not access the GreenFeed unit throughout the experiment except during period 3, resulting in no gaseous measurements in periods 1, 2, and 4 for this animal. Two additional cows failed to consistently access the GeenFeed unit during period 1 (cow 1) and period 2 (cow 2), resulting in missing gaseous data for these animals in their respective periods. Despite some missing data, we were able to obtain 91% ($n = 58$ observations) of the total number

Table 6. Apparent total-tract digestibility of nutrients and urinary excretion of nitrogenous compounds in lactating dairy cows fed low MP diets with different starch concentrations supplemented or not with rumen-protected Met, Lys, and His (RP-MLH)¹

Item	Diet ²				SEM	P-value ³		
	HS	HS+AA	RSPA	RSPA+AA		SC	MLH	SC × MLH
Apparent total-tract digestibility								
DM, %	66.7	66.8	66.9	67.0	0.52	0.61	0.79	0.96
OM, %	67.7	67.7	67.7	67.9	0.52	0.79	0.83	0.89
CP, %	66.4	67.5	68.1	68.2	0.63	0.03	0.26	0.35
NDF, %	42.4	42.1	48.9	49.5	1.30	<0.001	0.84	0.63
Starch, %	96.9	97.2	96.9	97.1	0.23	0.99	0.18	0.77
Ether extract, %	85.4	85.0	85.7	85.1	0.65	0.73	0.41	0.91
N intake, g/d	742	751	773	775	17.5	<0.001	0.34	0.52
Urinary volume and N excretion								
Creatinine, mM	6.96	6.88	5.57	5.56	0.21	<0.001	0.77	0.82
Volume, L/d	29.1	29.4	35.9	36.6	1.15	<0.001	0.58	0.84
Urea N, g/d	188	181	238	240	6.21	<0.001	0.58	0.25
Total N, g/d	281	273	337	334	7.20	<0.001	0.26	0.63
Urea N, % of total N	66.4	66.2	70.8	72.1	1.22	<0.001	0.62	0.50
Urea N, % of N intake	25.2	24.5	31.1	31.3	0.84	<0.001	0.60	0.37
Total N, % of N intake	38.2	37.0	44.1	43.7	0.95	<0.001	0.22	0.60

¹RP-MLH = 12 g/d of RP-Met (Smartamine M; Adisseo USA Inc.), 9 g/d of RP-Lys (AjiPro-L; Ajinomoto Health & Nutrition North America Inc.), and 15 g/d of RP-His (prototype supplement; Ajinomoto Co. Inc.).

²HS = high-starch diet; HS+AA = HS + RP-MLH; RSPA = reduced starch + palmitic acid-enriched supplement; and RSPA+AA = reduced starch + palmitic acid-enriched supplement + RP-MLH; BergaFat F100 (Berg+Schmidt America LLC), which is a ruminally stable lipid supplement containing 80% palmitic acid, was fed at 1.5% of the diet DM.

³SC = main effect of dietary starch concentration; MLH = main effect of RP-MLH supplementation; and SC × MLH = interaction between dietary starch concentration and RP-MLH supplementation.

of observations (n = 64) possible for enteric CH₄ production as shown in Supplemental Table S1.

RESULTS AND DISCUSSION

The amounts of digestible Met (7 g/d), Lys (2 g/d), and His (3 g/d) supplied by the RP-MLH supplements were originally calculated to meet the requirements of these 3 AA according to Schwab et al. (2005) using pre-experiment animal data reported in Supplemental Table S2 (https://figshare.com/articles/online_resource/Supplemental_Tables_S1-3/22422811; Zang, 2023). The nutrient and AA composition of feeds from previous studies done in our laboratory were also entered in the NRC (2001) software, except for the AA profile of grass hay for which NRC (2001) default values were used, to obtain estimates of digestible supply of dietary His, Met, and Lys (Supplemental Table S2). However, the supply of digestible Met from RP-Met was not sufficient to meet requirement in the formulated basal diets due to a mistake in our calculations. In addition to digestible Met, supply of digestible Lys and His was lower than that reported in the literature (Lee et al., 2012; Giallongo et al., 2016; Pereira et al., 2020; Zang et al., 2021b). Therefore, the general lack of effect of RP-MLH on response variables measured in the present study (Tables 5–8) or the absence of dietary starch concentration × RP-MLH supplementation interactions for most parameters evaluated was possibly associated

with the low supply of digestible AA from RP-MLH supplements. Furthermore, as ground corn was partially replaced with soyhulls and a palmitic acid-enriched lipid source in our experiment (Table 3), the impact of starch and palmitic acid on the variables tested cannot be isolated from each other and results should be interpreted considering this potential confounding effect.

Nutrient Composition of Feeds and Experimental Diets

Despite diets being formulated to be isonitrogenous, the CP concentration of RSPA was 0.9% unit greater than that of HS (Table 3) due to using a soyhulls source with an average CP concentration of 17.5% (Table 1), thus substantially greater than the mean CP values of 11.8% reported in the meta-analysis of Ipharraguerre and Clark (2003) and 13.5% in our previous research (Ghedini et al., 2018; Zang et al., 2021a,b). Although the CP concentration of the soyhulls fed herein was within the range of 9.4 to 19.2% compiled by Ipharraguerre and Clark (2003), the presence of contaminants resulting in nutrient composition changes cannot be excluded. For instance, the soyhulls ether extract concentration (mean = 7.05%; Table 1) was 60% greater than the highest ether extract content reported in Ipharraguerre and Clark (2003) paper. Compared with the NASEM (2021), the mean concentrations of CP and ether extract for the soyhulls used in the present study were

Table 7. Concentrations of plasma AA and carnosine in lactating dairy cows fed low MP diets with different starch concentrations supplemented or not with rumen-protected Met, Lys, and His (RP-MLH)¹

Item	Diet ²				SEM	<i>P</i> -value ³		
	HS	HS+AA	RSPA	RSPA+AA		SC	MLH	SC × MLH
EAA, μ <i>M</i>								
Arg	76.0	73.1	81.6	83.4	3.33	<0.01	0.86	0.41
His	56.0	57.5	49.8	56.4	3.02	0.06	0.04	0.18
Ile	122	114	134	130	5.56	<0.001	0.11	0.62
Leu	145	142	134	132	7.74	<0.01	0.56	0.99
Lys	80.8	78.8	83.7	88.0	4.17	0.08	0.74	0.35
Met	25.6	31.7	25.2	32.6	1.16	0.78	<0.001	0.55
Phe	49.2	49.2	46.5	47.9	1.71	0.09	0.54	0.54
Thr	101	90.6	104	102	4.28	0.05	0.08	0.27
Trp	49.9	48.1	48.8	49.2	1.55	0.99	0.57	0.37
Val	248	244	261	252	10.9	0.09	0.29	0.71
Total	953	930	975	974	35.3	0.13	0.58	0.61
NEAA, μ <i>M</i>								
Ala	262	261	279	283	13.3	0.02	0.90	0.79
Asn	48.3	43.6	48.5	48.6	1.80	0.15	0.19	0.19
Asp	3.44	3.32	3.23	3.30	0.17	0.41	0.85	0.51
Cit	88.1	82.3	100	101	5.06	<0.001	0.38	0.25
Cys	19.0	19.4	19.0	19.8	0.63	0.65	0.13	0.62
Gln	290	283	275	274	9.11	0.12	0.59	0.68
Glu	37.4	38.9	38.9	40.2	1.82	0.26	0.28	0.93
Gly	315	302	339	312	19.1	0.14	0.07	0.54
Orn	43.0	43.2	44.2	45.6	1.99	0.24	0.62	0.67
Pro	90.5	90.6	87.7	87.1	3.85	0.32	0.94	0.90
Ser	79.5	75.1	74.0	72.6	2.20	0.07	0.19	0.50
Taurine	36.6	39.1	33.1	38.6	1.56	0.08	<0.01	0.19
Tyr	51.8	48.9	50.8	51.8	2.78	0.56	0.54	0.20
Total	1,363	1,310	1,393	1,377	35.2	0.09	0.22	0.51
Carnosine, μ <i>M</i>	19.3	18.5	17.6	17.8	0.87	0.01	0.55	0.33

¹RP-MLH = 12 g/d of RP-Met (Smartamine M; Adisseo USA Inc.), 9 g/d of RP-Lys (AjiPro-L; Ajinomoto Health & Nutrition North America Inc.), and 15 g/d of RP-His (prototype supplement; Ajinomoto Co. Inc.).

²HS = high-starch diet; HS+AA = HS + RP-MLH; RSPA = reduced starch + palmitic acid-enriched supplement; and RSPA+AA = reduced starch + palmitic acid-enriched supplement + RP-MLH; BergaFat F100 (Berg+Schmidt America LLC), which is a ruminally stable lipid supplement containing 80% palmitic acid, was fed at 1.5% of the diet DM.

³SC = main effect of dietary starch concentration; MLH = main effect of RP-MLH supplementation; and SC × MLH = interaction between dietary starch concentration and RP-MLH supplementation.

1.5- and 3.7-fold greater, respectively, whereas the NDF concentration was 20% lower.

Dry Matter Intake and Milk Yield and Composition

Our study was designed as a 2 × 2 factorial arrangement of treatments with the goal to detect interactions between dietary starch concentration and RP-MLH supplementation. However, except for milk fat concentration (discussed below), no other significant interactions were observed (Tables 5–8). We are aware of only few studies investigating the interactions between dietary energy intake or starch concentration and RP-AA supplementation in lactating dairy cows. Rulquin and Delaby (1997) observed no interaction effects for DMI, yields of milk and milk components, and plasma concentration of EAA in cows fed diets containing low (i.e., 87% of the requirement) or normal (i.e., 100% of the requirement) energy level supplemented or not with RP-Met. Zang et al. (2021b) reported no dietary starch

concentration × RP-MLH supplementation interaction effects for DMI, milk yield and composition, apparent total-tract digestibility of nutrients, N and energy utilization, and most EAA with the exception of Arg and Lys. Note that Zang et al. (2021b) fed diets in which nonforage fiber sources (i.e., pelleted beet pulp and soyhulls) completely replaced ground corn at 30% of diet DM, resulting in a 22.1-percentage unit difference in starch concentration between high- and reduced-starch diets. Morris and Kononoff (2021), using a partially balanced incomplete block design study whereby dietary levels (DM basis) of fatty acids varied from 3.0 to 6.2%, starch from 20.2 to 31.3%, and digestible Lys from 0 to 17.8 g/d via RP-Lys supplementation, observed energy by digestible Lys interactions for milk protein concentration, tissue energy, and plasma Lys concentration. In general, the lack of interaction effects in the current study may have been associated with diets being moderately deficient in MP concentration (2.2%; Table 4) combined with low supply of digestible Met, Lys, and

Table 8. Gaseous emissions¹ and energy utilization and efficiency in lactating dairy cows fed low MP diets with different starch concentrations supplemented or not with rumen-protected Met, Lys, and His (RP-MLH)²

Item	Diet ³				SEM	<i>P</i> -value ⁴		
	HS	HS+AA	RSPA	RSPA+AA		SC	MLH	SC × MLH
Gaseous emission								
CO ₂ , kg/d	16.2	16.1	15.9	16.1	0.32	0.28	0.79	0.40
CH ₄ , g/d	519	523	527	536	22.2	0.48	0.67	0.85
CH ₄ , g/kg of DMI	18.5	18.1	18.6	19.0	0.68	0.37	0.90	0.46
CH ₄ , g/kg of ECM	11.8	11.8	11.4	11.4	0.42	0.24	0.99	0.86
Fraction, ⁵ Mcal/d								
GE intake	121	124	127	126	3.17	0.03	0.82	0.27
DE intake	83.9	86.4	89.7	88.1	2.46	0.02	0.79	0.22
ME intake	74.7	75.6	75.5	76.2	1.91	0.43	0.37	0.93
Component, ⁶ Mcal/d								
Fecal energy	37.5	37.4	37.6	37.6	1.02	0.75	0.99	0.98
Urinary energy	4.10	3.99	4.93	4.88	0.10	<0.001	0.27	0.64
CH ₄ energy	6.88	6.93	6.98	7.11	0.29	0.47	0.67	0.84
Milk energy	30.7	30.5	31.9	32.2	0.98	<0.001	0.88	0.49
Efficiency, %								
ME/DE	86.6	87.3	86.3	86.4	0.39	0.05	0.21	0.37
Milk energy/ME	41.5	41.2	42.5	42.6	0.91	0.03	0.70	0.36

¹Gaseous emissions were measured using the GreenFeed system (C-Lock Technology Inc.); data were derived from 8 individual spot measurements over a 4-d period.

²RP-MLH = 12 g/d of RP-Met (Smartamine M; Adisseo USA Inc.), 9 g/d of RP-Lys (AjiPro-L; Ajinomoto Health & Nutrition North America Inc.), and 15 g/d of RP-His (prototype supplement; Ajinomoto Co. Inc.).

³HS = high-starch diet; HS+AA = HS + RP-MLH; RSPA = reduced starch + palmitic acid-enriched supplement; and RSPA+AA = reduced starch + palmitic acid-enriched supplement + RP-MLH; BergaFat F100 (Berg+Schmidt America LLC), which is a ruminally stable lipid supplement containing 80% palmitic acid, was fed at 1.5% of the diet DM.

⁴SC = main effect of dietary starch concentration; MLH = main effect of RP-MLH supplementation; and SC × MLH = interaction between dietary starch concentration and RP-MLH supplementation.

⁵GE = gross energy; digestible energy (DE) intake (Mcal/d) = GE intake (Mcal/d) – fecal energy (Mcal/d); ME intake (Mcal/d) = DE intake (Mcal/d) – urinary energy (Mcal/d) – CH₄ energy (Mcal/d) (NRC, 2001).

⁶Urinary energy (Mcal/d) = [14.6 × urinary N output (g/d)]/1,000 (Morris et al., 2020); CH₄ energy (Mcal/d) = [CH₄ (L/d) × 9.45 (kcal/L)]/1,000; milk energy (Mcal/d) = [(0.0929 × milk fat%) + (0.0585 × milk true protein%) + (0.0395 × milk lactose%)] × milk yield (kg/d) (NASEM, 2021).

His from RP-MLH (7, 2, and 3 g/d, respectively; Table 4) and relatively similar NE_L concentration (Table 3).

Dry matter intake, milk yield and composition, feed efficiency, PUN concentration, and BCS and BW are shown in Table 5. We observed a dietary starch concentration by RP-MLH supplementation interaction ($P = 0.04$) for milk fat concentration. Specifically, milk fat concentration tended to decrease (–0.11% unit; $P = 0.08$) when RP-MLH was supplemented to HS cows, but no change (+0.07% unit; $P = 0.26$) was seen when it was fed to cows offered RSPA. An interaction tendency ($P = 0.07$) was also detected for milk yield, which was not affected (+0.4 kg/d; $P = 0.39$) by RP-MLH supplementation to HS, but it tended to decrease (–0.8 kg/d; $P = 0.08$) when RP-MLH was supplemented to RSPA. Therefore, it appears that both milk volume differences and RP-MLH supplementation were involved in the drop of milk fat concentration detected in the HS+AA diet. This is supported by the lack of dietary starch concentration × RP-MLH interaction ($P = 0.15$) on milk fat yield (Table 5).

Partially replacing ground corn with soyhulls and a palmitic acid-enriched supplement did not change ($P \geq 0.17$; Table 5) DMI, milk yield, or feed efficiency expressed as milk yield/DMI. Similarly, van Kneegsel et al. (2007) did not observe changes in DMI and milk yield in early-lactation dairy cows fed isocaloric diets formulated (DM basis) to be glucogenic (26.7% starch and 3.4% ether extract) or lipogenic (9.5% starch and 5.4% ether extract mostly from Ca salts of palm fatty acids and palm oil). In contrast, Boerman et al. (2015) reported that DMI tended to decrease (–0.5 kg/d) and milk yield decreased (–1.3 kg/d) in mid-lactation dairy cows fed a high-fiber, high-fat diet (50:50 forage:concentrate ratio; 16.1% starch) containing soyhulls plus a palmitic acid-enriched supplement versus a high-starch diet (40:60 forage:concentrate ratio; 32.5% starch) containing high-moisture corn and ground corn. Discrepancies in DMI and milk yield responses across studies may be associated with differences in dietary starch concentration and source, stage of lactation, production level, and types and amounts of nonforage fiber

sources and lipid supplements used in the diets. We also observed that in comparison to HS and HS+AA, feeding RSPA and RSPA+AA increased ($P < 0.001$) yields of 4% FCM (42.9 vs. 40.1 kg/d) and ECM (47.0 vs. 44.8 kg/d), and improved ($P < 0.001$) feed efficiency expressed as 4% FCM yield/DMI (1.50 vs. 1.40 kg/kg) or ECM yield/DMI (1.64 vs. 1.56 kg/kg). Boerman et al. (2015) showed that 3.5% FCM yield was 3.3% greater in the high-fiber, high-fat diet than in the high-starch counterpart, thus in agreement with the present study. Increased yields of 4% FCM and ECM seen in cows fed RSPA and RSPA+AA can be explained by improved milk fat yield because milk protein yield was not affected by diets as discussed in detail below.

Compared with HS and HS+AA, feeding RSPA and RSPA+AA increased ($P < 0.001$) the concentration (3.63 vs. 3.35%) and yield (1.65 vs. 1.50 kg/d) of milk fat (Table 5), which agree with van Knegsel et al. (2007) and Boerman et al. (2015) who fed lipogenic diets supplemented with Ca salts of palm fatty acids plus palm oil or palmitic acid, respectively. Increased milk fat yield in cows fed the RSPA and RSPA+AA diets was likely associated with more palmitic acid being incorporated into milk fat in the mammary gland (Loften et al., 2014). In addition, increased apparent total-tract digestibility of NDF with feeding RSPA and RSPA+AA versus HS and HS+AA (Table 6) possibly elevated the ruminal supply of acetate for de novo synthesis of fatty acids in mammary tissues. In fact, Ipharraguerre et al. (2002a,b) reported linear increases in milk fat yield and ruminal molar proportion of acetate in cows fed diets in which ground shelled corn was replaced by incremental amounts of soyhulls (up to 40% of the diet DM). Alternatively, decreased starch intake in RSPA and RSPA+AA relative to HS and HS+AA diets may have resulted in a more stable ruminal pH that shifted biohydrogenation pathways away from *trans*-10 18:1 and *trans*-10,*cis*-12 18:2 fatty acids, which are known to depress milk fat synthesis (Baumgard et al., 2002; Shingfield et al., 2009).

Milk true protein concentration was lower (3.10 vs. 3.17%; $P < 0.001$) in cows fed RSPA and RSPA+AA than in those receiving HS and HS+AA (Table 5), but milk true protein yield did not change ($P = 0.65$), indicating similar milk protein synthesis in mammary tissues. The effect of replacing starch with nonforage fiber sources on milk protein concentration and yield has not been consistent across studies. van Knegsel et al. (2007) reported that feeding glucogenic versus lipogenic diets did not affect milk protein concentration in dairy cows from wk 2 to 9 of lactation. In contrast, a diet by week interaction was observed for milk protein yield in the study of van Knegsel et al. (2007). However, they did not report in which week treatments differed and stated

that the interaction effect was small (van Knegsel et al., 2007). Boerman et al. (2015) reported that milk protein concentration and yield were both greater in mid-lactation cows fed a high-starch compared with a high-fiber, high-fat diet. Overall, differences in the type, amount, and processing of starch-based grains fed, type and levels of nonforage fiber sources included in the diets, and fatty acid profile of lipid supplements may explain the discrepancies in milk protein concentration and yield among van Knegsel et al. (2007), Boerman et al. (2015), and the present experiment. Feeding RSPA and RSPA+AA decreased (4.94 vs. 4.99%; $P < 0.01$) milk lactose concentration compared with HS and HS+AA diets (Table 5). However, milk lactose yield was not affected ($P = 0.53$; Table 5) by dietary starch concentration, thus indicating no change in milk lactose synthesis.

Milk N efficiency was lower (29.3 vs. 30.5%; $P < 0.01$) in cows fed RSPA and RSPA+AA than HS and HS+AA diets (Table 5). In addition, MUN (14.9 vs. 12.1 mg/dL) and PUN (15.6 vs. 12.3 mg/dL) concentrations both increased ($P < 0.001$) with feeding RSPA and RSPA+AA versus HS and HS+AA (Table 5). Decreased milk N efficiency and increased MUN and PUN concentrations were possibly associated with greater CP concentration in RSPA than HS diets (16.8 and 15.9%, respectively; Table 3) and consequent increase in N intake (774 vs. 747 g/d; Table 6). Olmos Colmenero and Broderick (2006) observed a linear decrease in milk N efficiency (from 36.5 to 25.4%) and linear increases in MUN (from 7.7 to 15.6 mg/dL) and BUN (from 10.7 to 24 mg/dL) in dairy cows fed diets with incremental levels of CP (from 13.5 to 19.4%) and increased N intake (from 483 to 710 g/d). Nousiainen et al. (2004) reported a positive linear relationship [$y = 0.17 \pm 0.005x - 14.2 \pm 0.849$; $R^2 = 0.778$; $n = 306$ observations] between dietary CP concentration (g/kg of DM) and MUN (mg/dL). Based on Nousiainen et al. (2004) regression equation, the predicted difference in MUN concentration between our HS and RSPA basal diets was 1.53 mg/dL. However, the actual difference in MUN concentration for HS and RSPA averaged 2.7 mg/dL or 76.5% greater than predicted, suggesting that factors beyond dietary CP level could have been involved. Dietary starch concentration averaged (DM basis) 32.6 and 21.7% for HS and RSPA, respectively (Table 3), implying that less ruminally fermentable energy was available for microbial growth when feeding RSPA. This combined with greater RDP supply (+118 g/d; Table 4) and N intake (+27 g/d; Table 6) in RSPA and RSPA+AA versus HS and HS+AA may have lowered the efficiency in which ruminal microorganisms used $\text{NH}_3\text{-N}$ for microbial protein synthesis, with excess NH_3 being absorbed through ruminal epithelium

and converted to urea in the liver. Ipharraguerre et al. (2002b) showed a linear increase in ruminal $\text{NH}_3\text{-N}$ concentration (from 12.6 to 17.8 mg/dL) when substituting ground shelled corn with incremental amounts of soyhulls despite no change in MUN concentration (Ipharraguerre et al., 2002a). In contrast, Gao and Oba (2016) demonstrated that the concentrations of ruminal $\text{NH}_3\text{-N}$ and MUN both decreased in dairy cows fed high-NFC diets where rolled corn grain, sucrose, or lactose replaced beet pulp. Milk SCC, BCS, BW, and changes (i.e., gain) in BCS and BW were not affected ($P \geq 0.15$) by diets (Table 5).

Supplementation with RP-MLH did not affect ($P \geq 0.52$) DMI and yields of milk, 4% FCM, and ECM (Table 5). Likewise, supplemental RP-MLH had no effect ($P \geq 0.20$) on feed and milk N efficiency, milk component yields, milk SCC, concentrations of milk lactose, MUN, and PUN, BW, and BCS and BCS change (Table 5). Milk true protein concentration ($P = 0.06$) and BW gain ($P = 0.09$) tended to be greater in cows fed HS+AA and RSPA+AA than HS and RSPA (Table 5). Despite the tendency for increased milk true protein concentration with feeding HS+AA and RSPA+AA, milk true protein yield did not differ across diets, indicating similar milk protein synthesis. It is important to note that the amounts of digestible Met, Lys, and His (7, 2, and 3 g/d respectively) supplied through the RP-MLH supplements were substantially lower compared with those from previous experiments (Lee et al., 2012; Giallongo et al., 2016; Pereira et al., 2020; Zang et al., 2021b), which may have limited our capacity to detect a potential effect of RP-MLH on milk protein synthesis. However, Zang et al. (2021b) supplementing greater levels of digestible Met, Lys, and His (15, 16, and 7 g/d, respectively) to diets more MP deficient (mean = 4.3%) than those used in the current experiment also observed no effect of RP-MLH on milk protein yield. This agrees with Pereira et al. (2020) who supplemented 16 g/d of digestible Met via RP-Met to cows receiving diets with negative MP balance ranging from -71 to -340 g/d. In contrast, when digestible Met, Lys, and His were supplied by RP-MLH (18, 24, and 12 g/d, respectively; Lee et al., 2012) or jugular infusion (21, 38, and 20 g/d, respectively; Yoder et al., 2020) to diets averaging 14.3% (Lee et al., 2012) and 15% (Yoder et al., 2020) MP deficiency, milk protein yield increased. Although these results suggest that milk protein synthesis is more responsive to AA supplementation in situations where MP deficiency is pronounced (~14–15%), this has not been always the case. For instance, Giallongo et al. (2016) observed increased milk protein yield in cows supplemented with 18, 28, and 9 g/d of digestible Met, Lys, and His, respectively, in diets averaging 2% MP deficiency, thus indicating a direct RP-MLH effect.

Nutrient Digestibility and Urinary Excretion of Nitrogenous Metabolites

Apparent total-tract digestibility of nutrients, N intake, and urinary excretion of nitrogenous metabolites are presented in Table 6. Nitrogen balance results are shown in Supplemental Table S3 (https://figshare.com/articles/online_resource/Supplemental_Tables_S1-3/22422811; Zang, 2023) and are not discussed in this paper. No dietary starch concentration by RP-MLH supplementation interactions ($P \geq 0.25$) were observed for N intake, apparent total-tract digestibility of nutrients, urinary concentration of creatinine, and urinary excretion of nitrogenous metabolites (Table 6).

Apparent total-tract digestibilities of DM, OM, starch, and ether extract were not affected ($P \geq 0.61$) by dietary starch concentration (Table 6). Zang et al. (2021b) also reported no effect of dietary starch concentration on the apparent total-tract digestibilities of DM and OM. In contrast, total-tract digestibilities of CP (68.2 vs. 67.0%) and NDF (49.2 vs. 42.3%) were greater ($P \leq 0.03$) in cows fed RSPA and RSPA+AA than HS and HS+AA (Table 6). Improved CP digestibility may have been caused by increased dilution of fecal metabolic N due to greater N intake (+27 g/d; Table 6) with feeding RSPA and RSPA+AA versus HS and HS+AA. We also observed that NDF digestibility in the total-tract increased by 16.3% in RSPA and RSPA+AA compared with HS and HS+AA, corroborating results of Zang et al. (2021b) and Boerman et al. (2015). Increased fiber digestibility may be linked to the high extent of NDF digestion typically seen with nonforage fiber sources (Firkins, 1997), elevated ruminal pH (Sarwar et al., 1992; Ipharraguerre and Clark, 2003), or both (Firkins, 1997; Ipharraguerre and Clark, 2003).

Cows fed RSPA and RSPA+AA had lower ($P < 0.001$) urinary creatinine concentration (5.57 vs. 6.92 mM) and greater ($P < 0.001$) urinary volume (36.3 vs. 29.3 L/d) than those fed HS and HS+AA (Table 6). Nitrogen intake increased ($P < 0.001$) by 3.6% (774 vs. 747 g/d) with feeding RSPA and RSPA+AA versus HS and HS+AA, thus in line with the observed increase in urinary production needed for excreting excess N. Moreover, K intake increased ($P < 0.001$) by 13.8% (371 vs. 326 g/d; data not shown) when comparing RSPA and RSPA+AA with HS and HS+AA, which may further explain increased urinary volume (Bannink et al., 1999; Eriksson and Rustas, 2014).

Urinary excretion of urea N (239 vs. 185 g/d) and total N (336 vs. 277 g/d) were greater ($P < 0.001$) in cows fed RSPA and RSPA+AA than HS and HS+AA (Table 6). Urinary excretion of urea N, expressed as a proportion of total urinary N (71.5 vs. 66.3%) or N in-

take (31.2 vs. 24.9%), was also greater ($P < 0.001$) with feeding RSPA and RSPA+AA versus HS and HS+AA. Similarly, urinary excretion of total N, expressed as a proportion of N intake (43.9 vs. 37.6%), increased ($P < 0.001$) in RSPA and RSPA+AA compared with HS and HS+AA. In agreement with our results, urinary N excretion (g/d or % of N intake) increased in lactating dairy cows offered a high-fat diet rather than a high-starch diet (Morris et al., 2020). Decreased starch intake can limit microbial protein synthesis in the rumen and milk protein yield, ultimately increasing urinary N excretion (Morris et al., 2020; Zang et al., 2021b).

Supplementation with RP-MLH did not affect ($P \geq 0.18$) the apparent total-tract digestibility of nutrients, N intake, and urinary volume and excretion of nitrogenous compounds (Table 6). These results generally agree with those from our previous study in which cows were also fed diets with 2 levels of starch supplemented or not with RP-MLH (Zang et al., 2021b).

Plasma Concentrations of AA and Carnosine

Plasma concentrations of AA and the His-containing peptide carnosine are shown in Table 7. No significant interactions were observed between dietary starch level and RP-MLH supplementation for the concentrations of EAA, NEAA, and carnosine in plasma of lactating dairy cows. In our previous research (Zang et al., 2021b), significant dietary starch concentration and RP-MLH supplementation interactions were found for the plasma concentrations of Arg, Lys, and Orn, which increased when RP-MLH was supplemented to the reduced-starch diet but not the high-starch counterpart. Rulquin and Delaby (1997) reported no energy intake $\times$ RP-Met interaction for the plasma concentrations of any EAA and NEAA measured in their experiment. As discussed earlier, the lack of interaction effects herein is possibly linked to the moderate deficiency in MP concentration of the basal diets and low supply of digestible Met, Lys, and His from RP-MLH.

Feeding RSPA and RSPA+AA versus HS and HS+AA increased ($P \leq 0.05$) the plasma concentrations of Arg, Ile, and Thr, and decreased ($P < 0.01$) Leu concentration (Table 7). Furthermore, the plasma concentrations of His and Phe tended ($P \leq 0.09$) to be lower, whereas those of Lys and Val tended ($P \leq 0.09$) to be greater in cows fed RSPA and RSPA+AA than HS and HS+AA diets (Table 7). In contrast, dietary starch concentration did not affect ($P \geq 0.13$) the plasma concentrations of Met, Trp, and total EAA. These changes in the plasma concentrations of EAA may be associated with postabsorptive processes including increased or decreased utilization of EAA by

hepatic and mammary tissues (Lapierre et al., 2006) as the duodenal flows of nonammonia, nonmicrobial N, and microbial NAN were not affected when ground shelled corn was replaced with incremental amounts of soyhulls (Ipharraguerre et al., 2002b).

Zang et al. (2021b) reported that cows fed the high-starch diet had lower plasma concentrations of most EAA than those fed the reduced-starch counterpart, partially agreeing with the present results (Table 7). Increased mammary extraction of EAA for synthesis of milk protein may have lowered the concentration of EAA in plasma (NASEM, 2021) in the study of Zang et al. (2021b). In fact, milk protein yield increased by 8.4% with feeding high versus reduced-starch diets in the experiment of Zang et al. (2021b). The difference in starch intake between the high and reduced-starch diets averaged 4.89 kg/d in Zang et al. (2021b) and 3.09 kg/d herein (Table 5), which may further explain the discrepancies seen in the plasma concentrations of EAA comparing these 2 studies. Rius et al. (2010) observed that the plasma concentrations of Ile, His, Lys, Phe, and Val decreased in feed-restricted dairy cows abomasally infused with starch and starch plus casein versus water and casein. It is worth noting that according to Lapierre et al. (2006), plasma concentration of EAA can be affected by many other factors including liver extraction and catabolism in the portal-drained viscera and extra-hepatic tissues as discussed previously.

Cows fed RSPA and RSPA+AA had increased ($P \leq 0.02$) plasma concentrations of Ala and Cit, and tended to have increased ($P < 0.09$) total NEAA and decreased ($P \leq 0.08$) Ser and taurine compared with those fed HS and HS+AA (Table 7). Contrarily, feeding RSPA and RSPA+AA versus HS and HS+AA did not affect ($P \geq 0.12$) the plasma concentrations of all remaining NEAA. Elevated plasma concentration of Arg and Cit coincided with increased MUN and PUN concentrations, suggesting that a greater pool of urea cycle AA was available for hepatic ureagenesis in RSPA and RSPA+AA than HS and HS+AA diets.

The plasma concentration of carnosine decreased ($P = 0.01$) with feeding RSPA and RSPA+AA versus HS and HS+AA (Table 7). According to Maynard et al. (2001) and Boldyrev et al. (2013), carnosine is present almost exclusively in skeletal muscles as a dipeptide synthesized from His and Ala via carnosine synthase. However, there is evidence that carnosine is transported from muscles to plasma due to presence of mRNA transcripts for the peptide/His transporter 1 and 2 in skeletal muscle of mice and humans (Boldyrev et al., 2013; Everaert et al., 2013). Reduced plasma concentration of carnosine in cows fed RSPA and RSPA+AA is difficult to explain, but carnosine could have been used as an

endogenous source of His (Lapierre et al., 2008; Lapierre et al., 2021) or transported less efficiently from muscle to plasma due to unknown mechanisms.

Supplementation with RP-MLH did not affect ($P \geq 0.11$) the plasma concentrations of most EAA and NEAA as shown in Table 7. However, the plasma concentrations of His and Met increased ($P \leq 0.04$) in cows receiving RP-MLH supplementation, indicating that the RP-His and RP-Met supplements provided additional digestible His and Met, respectively. In contrast, the plasma concentration of Lys was not changed in response to supplemental RP-MLH, probably due to limited supply of digestible Lys via RP-Lys (Table 4). The plasma concentration of Gly tended to be lower ($P = 0.07$) with feeding HS+AA and RSPA+AA versus HS and RSPA, whereas that of taurine increased ($P < 0.01$) by 11.5% (38.9 vs. 34.9 μM) with RP-MLH supplementation (Table 7). It is well known that taurine is synthesized via sequential oxidation and decarboxylation reactions using Cys, which, in turn, originates from transsulfuration of Met as reviewed by Baker (2006) and Ripps and Shen (2012). Therefore, increased plasma concentration of Met in cows supplemented with RP-MLH is in line with the observed increase in circulating taurine despite no change ($P = 0.13$) in Cys concentration (Table 7). Zang et al. (2021b) reported increases in plasma concentrations of Met (+68.9%), Cys (+8.2%), and taurine (+23.3%) in dairy cows receiving supplemental RP-MLH.

Gaseous Emissions and Energy Utilization

Emissions of CO_2 and enteric CH_4 and dietary energy intake, utilization, and efficiency are presented in Table 8. No interactions ($P \geq 0.22$) between dietary starch concentration and RP-MLH supplementation were observed for any of the energy-related variables, which agree with Zang et al. (2021b).

Dietary starch concentration did not significantly affect CO_2 production, as well as enteric CH_4 production, CH_4 yield, and CH_4 intensity (Table 8). In our previous research, cows fed the high-starch diet had lower enteric CH_4 production (434 vs. 545 g/d), CH_4 yield (17.3 vs. 21.2 g/kg of DMI), and CH_4 intensity (10.7 vs. 13.6 g/kg of ECM) than those fed the reduced-starch diet. It is well established that enteric CH_4 production is a function of DMI (Niu et al., 2018). Dry matter intake did not change in the present study (Table 5), which is in line with the lack of treatment effect on enteric CH_4 production. Contrarily, DMI and enteric CH_4 production increased in Zang et al. (2021b) experiment, thus clarifying the discrepancy in CH_4 emissions between our 2 studies. Niu et al. (2018) also reported that prediction of enteric CH_4 production improved when

dietary NDF concentration was included in the models together with DMI. The difference in NDF intake (data not shown) between the high- and reduced-starch diets averaged 2.37 and 1.41 kg/d in Zang et al. (2021b) and the present study, respectively, further explaining the different responses in enteric CH_4 production in these experiments.

Cows fed RSPA and RSPA+AA had greater ($P \leq 0.03$) intake of GE and DE than those fed HS and HS+AA diets (Table 8). In contrast, dietary starch concentration did not affect ($P \geq 0.43$) ME intake, fecal energy, and CH_4 energy (Table 8). Feeding RSPA and RSPA+AA increased ($P < 0.001$) urinary energy output, thus in line with the effect of diets on urinary N excretion (Table 6) discussed earlier. The efficiency of converting DE into ME decreased ($P = 0.05$) slightly in cows fed RSPA and RSPA+AA versus HS and HS+AA possibly in response to increased urinary energy output as CH_4 energy did not change across diets (Table 8). Contrarily, energy efficiency, expressed as milk energy/ME, improved ($P = 0.03$) with feeding RSPA and RSPA+AA versus HS and HS+AA (Table 8). Recently, we observed that feeding high-starch diets improved ME/DE and tended to improve milk energy/ME compared with reduced-starch diets (Zang et al., 2021b). A major difference between the current study and Zang et al. (2021b) experiment was palmitic acid supplementation. Therefore, these results indicate that palmitic acid improved energy use efficiency in dairy cows through (1) better conversion of ME into milk energy and (2) increased milk energy output. van Knegsel et al. (2007) and Boerman et al. (2015) reported increased milk energy in cows supplemented with Ca salts of palm fatty acids plus palm oil or palmitic acid, respectively.

Supplementation with RP-MLH did not affect ($P \geq 0.21$) gaseous emissions or any energy utilization variables (Table 8). These results are expected because supplemental RP-MLH had no effect on production performance and apparent total-tract digestibility of nutrients. Similarly, Zang et al. (2021b) reported no change in enteric CH_4 emissions, as well as energy utilization and efficiency in dairy cows fed low MP diets supplemented with RP-MLH.

CONCLUSIONS

We designed this experiment to test the hypothesis that the partial substitution of ground corn with soy-hulls and a palmitic acid-enriched lipid source could interact with RP-MLH to modulate milk yield and composition and nutrient utilization in dairy cows fed low MP diets. However, apart from milk fat concentration, no other interactions were observed. Note that the

amounts of digestible Met (7 g/d), Lys (2 g/d), and His (3 g/d) supplied via RP-MLH were lower than what have been reported in the literature and our results should be interpreted within this context. Compared with HS and HS+AA, cows fed RSPA and RSPA+AA had greater yields of milk fat, 4% FCM, and ECM, and improved feed efficiency (i.e., 4% FCM yield/DMI and ECM yield/DMI) and apparent total-tract digestibilities of CP and NDF. In contrast, feeding RSPA and RSPA+AA decreased milk N efficiency and increased urinary excretion of urea N and total N. Cows fed RSPA and RSPA+AA were more efficient in converting ME into milk energy than those fed HS and HS+AA. Supplementation with RP-MLH increased the plasma concentrations of Met and His but did not affect production performance and N and energy utilization.

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